Family Violence in Victoria

Data Visualisation Project Proposal

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Contents

1 Project description and motivation **1**

2 Project value: Research questions **1**

3 Dataset details: Data collection and wrangling **2**

4 Project plan and timeline **4**

5 References **4**

1 Project description and motivation

For this project, I will be investigating family violence in Victoria. Family violence is far too prevalent and too often goes unnoticed. Data reporting and visualisation can help organisations to better support those experiencing family violence, and can help policy makers to reduce incidence of family violence.

Intended audience: Local and state government policy makers, along with family violence support services.

Purpose: To support the target audience in understanding who is most affected by family violence, as well as understanding how family violence has changed over time, and whether these changes correlate with changes in legislation. This will assist decision-making around family violence prevention, as well as allowing support services to tailor their support to the demographics and locations most affected.

2 Project value: Research questions

Do LGAs with higher rates of family violence share any common characteristics which LGAs with lower rates of family violence don't possess? For example, lower socio-economic index, higher unemployment rate, higher gambling expenditure, median rent etc.

How has family violence in Victoria changed over time?

Which demographics are most affected by family violence, and does this differ from non-FV crime (specifically non-FV crimes against people)?

3 Dataset details: Data collection and wrangling

I obtained a number of datasets from the Victoria open data website, and the Crime Statistics Agency website. The first datasets I obtained contain the number of family incidents attended to by Victoria Police

in each local government area (LGA) from 2015 to 2020, and from 2021 to 2025. These were obtained from the following urls:

https://files.crimestatistics.vic.gov.au/2021-07/Data_Tables_LGA_Family_Incidents_Year_Ending_September_2020.xlsx

https://files.crimestatistics.vic.gov.au/2025-12/Data_Tables_LGA_Family_Incidents_Year_Ending_September_2025.xlsx

Each of the above excel workbooks contain a number of independent spreadsheets. The three sheets that I will be using for this project contain the number of family incidents for each LGA, the number of affected family members (AFMs) by age group and LGA, and the number of affected family members by sex and LGA. I exported each sheet as a CSV file, and first merged each pair of datasets so that the full timespan from 2015 to 2025 were represented in each dataset. This gave the following three datasets: family_incidents, afm_age, and afm_sex (CSA 2026b).

The raw family_incidents dataset had 6 columns and 870 rows, however, I removed three columns to leave only columns representing the year, LGA, and rate of family incident per 100,000 people. I then converted the rate of incident column to a numeric datatype, leaving the year column as an integer type and the LGA column as a character type. This dataset also contained group totals for each police region, along with justice and immigration facilities and unincorporated areas. I removed these rows from the dataset to leave 790 rows and 3 columns.

The raw afm_age dataset had 5 columns and 4726 rows. I removed one column, leaving columns for year, LGA, age group, and number of affected family members. Next, I converted the number of AFMs column to an integer datatype before creating a new column for each distinct value in the age group column. This left a dataset with 8 columns and 790 rows.

The raw afm_sex dataset had 5 columns and 1580 rows. I removed one column, leaving columns for year, LGA, sex, and number of AFMs. Next, I converted the number of AFMs column to an integer datatype before creating a new column for each distinct value in the sex column. This left a dataset with 4 columns and 790 rows (CSA 2026b).

Next, I obtained data regarding the number of offences recorded by Victoria Police from 2015 to 2025. This dataset can be found at the following link:

https://files.crimestatistics.vic.gov.au/2025-12/Data_Tables_LGA_Recorded_Offences_Year_Ending_September_2025.xlsx

Again, this workbook contained multiple sheets, however, I have decided to use only one of them, which holds records of offences by offence type, LGA, and year. I exported this dataset as a CSV file. This raw criminal_offence dataset contained 10 columns and 52825 rows. I removed 5 columns to leave columns for year, LGA, offence division, offence subgroup, and offence rate per 100,000 people for each LGA. Next, I removed all rows where the offence division was not 'crimes against the person', as I am interested in comparing rates of FV crimes to rates of similar non-FV crimes. I then removed the offence division column, and removed all rows where the criminal offence was not categorised as either FV related or non-FV related. Finally, I converted the offence rate to a numeric datatype, and reformatted the offence subgroup values to remove the indexing code, leaving only the name of the offence. I then created a new column for each distinct offence subgroup, with the values of these columns coming from the offence rate column. This final dataset has 12 columns and 790 rows (CSA 2026b).

I also sourced datasets containing data regarding gambling expenditure, SEIFA scores, the number of venues and gaming machines, and the unemployment rate for each LGA, from the Victorian Gambling and Casino Control Commission, through the Data Vic website. The data for 2021 to 2024 can be found at the following link:

https://www.vgccc.vic.gov.au/sites/default/files/2024-12/yearly_density_statistical_release_nov_24-%282%29.xlsx

The data for 2007 to 2021 can be found at the following link:

https://www.vgccc.vic.gov.au/sites/default/files/2024-12/historical_population_density_stats_egm_data-%281%29.xls

For both of these datasets, data for each year is held in a separate spreadsheet, so the first step in preparing this data was to export each sheet as a CSV. Since the above datasets covered the year ending 2016 to the year ending 2025, I only used sheets covering these years. Thus, I had 9 CSVs, as data was not available for the year ending 2025 (State Government of Victoria 2025a; State Government of Victoria 2025b).

This data was formatted in excel to be easily readable for humans. Therefore, when I read them into R using the usual methods, I was left with dataframes that contained many columns and rows filled with null values. To deal with this, I skipped the first few rows (the exact number of which differed for each CSV file), and defined the number of rows to read in as 79, and the number of columns as 19. I also manually defined the column names, as the CSV file contained some names in one row and some in another.

Once I had successfully read in the files, I added a year column, and concatenated the datasets into a single dataset with 711 rows and 19 columns. I removed 11 columns, leaving a column for year, LGA, SEIFA disadvantage score, SEIFA advantage/disadvantage score, number of adults per gaming venue, number of electronic gaming machines per 1000 adults, gambling expenditure per adult, and unemployment rate. I then converted the LGA names to the same format as the previous datasets, and ensured that all numerical columns were set to numeric datatypes.

The last datasets I sourced came from Homes Victoria (via the Data Vic website). One dataset contains quarterly median rents for each LGA, found at the following url:

`https://www.dffh.vic.gov.au/quarterly-median-rents-local-government-area-september-quarter-`

I exported this dataset as a CSV, and the raw data had 215 columns and 88 rows. Each quarter of a year had two columns assigned to it, representing the number of properties for rent and the median rent. As the other datasets I used had the year ending in September, I wanted to use the median rents for September. Therefore, I removed columns representing quarters other than September. I also removed columns representing the number of properties for rent, as I am only interested in the median rent. I reformatted the column headings so that they contained only the year (not the month), and created a year column. Finally, I removed rows representing years before 2016. This left me with a dataset consisting of 790 rows and 3 columns (State Government of Victoria 2026b).

The other dataset of interest contains the percentage of rentals which were classed as affordable in each LGA, found at this link:

`https://www.dffh.vic.gov.au/affordable-lettings-local-government-area-september-quarter-20`

This dataset had the same properties as that for median rents, and therefore I followed the same data wrangling procedure. The final dataset contained 790 rows, with 3 columns representing year, LGA, and percentage of properties for rent classed as affordable (State Government of Victoria 2026a).

I used a full join on LGA and year to combine all of the above datasets. This final dataset has 711 rows and 29 columns.

I also requested a dataset containing LGA boundary polygons as a shapefile, through Victorian government Department of Energy, Environment and Climate Action Spatial DataShare website. I have not yet received this dataset, so I have not done any data wrangling on this dataset (State Government of Victoria 2026c).

4 Project plan and timeline

Week	Milestone to reach	Tasks to complete to reach milestone
Week 2	Develop research questions Finish data wrangling	Find datasets Explore possible topics Complete data cleansing and join datasets
Week 3	Complete exploratory data analysis	Create basic plots to investigate relationships between variables Check that research questions are able to be answered with available data and make changes if necessary
Week 4	Complete 5dS	Come up with design ideas to complete 5dS Check that the final design sheet is realistic and you are able to produce it with the software available
Week 5	Build visualisation Write report	Convert design ideas to code

5 References

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